Routing inside the ansatz: connectivity-aware adaptive variational quantum dynamics on square and heavy-hex qubit lattices

Anonymous Author(s)

Abstract

Adaptive variational quantum dynamics simulation (AVQDS) builds compact circuits for real-time evolution by growing a pseudo-Trotter ansatz whenever McLachlan's distance exceeds a threshold, but its two-qubit gates ignore which qubits are physically coupled. We show that the trial step deciding each growth event is highly degenerate. In unconstrained runs 65–94% of the decisions are *terminal*, i.e. several operators would restore the accuracy threshold, and an admissible single-qubit alternative exists for 76–91% of the appended two-qubit rotations. We use this freedom to make ansatz construction device-aware. Connectivity-aware AVQDS (CA-AVQDS) leaves the equations of motion, regularisation, step control, pool and accuracy threshold unchanged. It tracks the layout of the growing circuit, appends among admissible candidates the one with the lowest compiled CNOT cost, and routes the rest with persistent SWAP chains chosen by a McLachlan-weighted look-ahead. On the 8-site mixed-field Ising ring and Lieb–Schultz–Mattis chain, every compiled circuit is compliant and every final fidelity stays within 0.0014 of AVQDS. Hyper-parameters were fixed on a disjoint development set. Over five tie-breaking orders, the mean CNOT multiplier relative to unconstrained AVQDS is 0.73 on a 2×4 lattice and 1.32 on a heavy-hex subgraph, against 2.04 for the strongest shortest-path router and 1.42 for SABRE; depth follows the same order. On an unseen linear chain the multiplier is 1.15 (naive 1.65, SABRE 1.41). An exact factorisation separates a topology-independent substitution gain from the routing overhead. It shows that CA-AVQDS circuits are cheaper under any router and identifies our online router as the bottleneck for the ring on heavy-hex, where SABRE re-routing of CA-AVQDS circuits reaches 1.18.

1 Introduction

Simulating the real-time dynamics of interacting quantum systems is a natural task for quantum computers [1–3]. On near-term hardware [4, 5], the depth of Trotter circuits [6–8] grows linearly with the simulated time and soon exceeds the coherence budget. This has motivated variational alternatives that evolve a parametrised circuit by a variational principle [9–13], by successive projections [14, 15], or by compressing Trotter circuits [16–20]. Adaptive variational quantum dynamics simulation (AVQDS) [21] grows a pseudo-Trotter ansatz from a pool of Pauli rotations whenever McLachlan's distance exceeds a threshold. It reaches high fidelity with up to two orders of magnitude fewer CNOT gates than first-order Trotterisation, and the idea has been carried over to imaginary time, thermal states, open systems and layered circuits [22–25], in the spirit of ADAPT-type ground-state methods [26–29].

These resource counts assume that any two qubits can interact. Superconducting processors realise sparse coupling graphs, such as square lattices [30] or the heavy-hex lattice [31, 32]. A two-qubit gate between uncoupled qubits must then be routed with SWAP gates, three CNOTs each [33, 34], and compilers that minimise this overhead form a large literature of their own [35–41]. For AVQDS

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the mismatch is structural. The two-qubit generators of its Hamiltonian pool sit on the bonds of the model's interaction graph (a ring for the periodic mixed-field Ising model, a chain for the Lieb–Schultz–Mattis model), and whether all bonds can sit on device edges is a Hamiltonian-cycle question. On a 2×4 lattice they can. On an 8-qubit heavy-hex subgraph, a tree, at least three ring bonds and two chain bonds stay non-adjacent under every placement (Lemma 1), so any compliant implementation must move qubits during the simulation.

The standard response is to compile the finished circuit. We argue that an adaptive ansatz offers a better entry point, because its circuit is not given: it is being decided, and the decision is strongly underdetermined. In unconstrained AVQDS runs on the benchmarks studied here, 65–94% of all growth decisions are *terminal*: the appended operator restores the accuracy threshold, and typically four to six operators would have done so. Whenever AVQDS appends a two-qubit rotation, an admissible single-qubit rotation (in the sense of Eq. (4) below) exists in 76–91% of the cases. The original algorithm resolves this freedom with a tie-break that carries no physical information and, as we show, is decided by floating-point round-off.

This paper introduces *connectivity-aware AVQDS* (CA-AVQDS), which spends this freedom on the device instead. Our contributions are:

- A device layer for adaptive variational dynamics (Sec. 2). It leaves the McLachlan equations of motion, regularisation, step control, pool and accuracy threshold untouched and reads only the trial distances the algorithm computes anyway. It consists of a lexicographic selection rule (admissibility first, compiled CNOT cost on the device second, the original rule third), persistent SWAP routing with a McLachlan-weighted look-ahead, and an exact commutation-aware placement that we evaluate but do not select. The circuits are compliant by construction, the AVQDS accuracy criterion is kept, and the method reduces to AVQDS when costs tie.
- A structural analysis of when routing is unavoidable (Lemma 1) and a quantitative account of the degeneracy of the McLachlan trial step that makes ansatz-level routing possible.
- A paired, five-seed evaluation on the original AVQDS benchmarks and topologies A (2×4 lattice) and B (heavy-hex), plus an out-of-sample linear chain C (Secs. 4–5). All hyper-parameters were fixed beforehand on a disjoint development set. CA-AVQDS keeps the final fidelity within 0.0014 of AVQDS on every run. Its mean CNOT multipliers relative to unconstrained AVQDS are 0.73 on A, 1.32 on B and 1.15 on C. For comparison, the strongest naive router gives 2.04 (B) and 1.65 (C), and SABRE applied to the AVQDS circuits gives 1.42 (B) and 1.41 (C). An exact factorisation separates a topology-independent substitution gain from the topology-induced overhead, and ablations, failure and cost analyses identify where the method helps and where it does not.

2 Methodology

2.1 Adaptive McLachlan dynamics (held fixed)

The variational state is a pseudo-Trotter product of Pauli rotations acting on a reference state,

$$|\Psi[\boldsymbol{\theta}]\rangle = \prod_{\mu=0}^{N_\theta-1} e^{-i\theta_\mu \hat{A}_\mu} |\Psi_0\rangle, \tag{1}$$

where every generator $\hat{A}_\mu$ is a Pauli string taken from an operator pool $\mathcal{P}$ and the product is ordered with $\mu = 0$ acting first. McLachlan's principle [9, 11] minimises the distance between the variational and the exact evolution of $\rho = |\Psi\rangle\langle\Psi|$,

$$L^2 = \Big\| \sum_\mu \partial_{\theta_\mu}\rho\, \dot{\theta}_\mu - \mathcal{L}[\rho] \Big\|_F^2 = \dot{\boldsymbol{\theta}}^\top M \dot{\boldsymbol{\theta}} - 2V^\top \dot{\boldsymbol{\theta}} + 2\,\mathrm{var}_{\boldsymbol{\theta}}(\hat{H}), \tag{2}$$

with $M_{\mu\nu} = 2\,\mathrm{Re}\big[\langle\partial_\mu\Psi|\partial_\nu\Psi\rangle + \langle\partial_\mu\Psi|\Psi\rangle\langle\partial_\nu\Psi|\Psi\rangle\big]$ and $V_\mu = 2\,\mathrm{Im}\big[\langle\partial_\mu\Psi|\hat{H}|\Psi\rangle + \langle\Psi|\partial_\mu\Psi\rangle\langle\hat{H}\rangle\big]$. The equations of motion $\dot{\boldsymbol{\theta}} = (M + \xi\,\mathbb{I})^+ V$ use Tikhonov regularisation $\xi = 10^{-6}$ [42], the optimal distance is $L^2 = 2\,\mathrm{var}(\hat{H}) - V^\top (M + \xi\,\mathbb{I})^+ V$, and parameters are advanced by forward Euler with the adaptive step $\delta t = \min(\delta t_{\max}, \delta\theta_{\max}/\max_\mu |\dot{\theta}_\mu|)$, $\delta t_{\max} = 0.05$, $\delta\theta_{\max} = 0.005$ [21]. Whenever $L^2 \geq L^2_{\mathrm{cut}} = 10^{-3}$ the ansatz is grown by the *trial-append loop*: for every $\hat{A}_\nu \in \mathcal{P}$

(except the operator appended last) the circuit is extended by $e^{-i\theta' \hat{A}_\nu}$ at $\theta' = 0$, the new row of M and entry of V are evaluated, and the trial distance L^2_ν is computed. The original rule appends the operator with the smallest L^2_ν; among candidates that already satisfy $L^2_\nu < L^2_{\mathrm{cut}}$ it prefers the one with the smaller $\max_\mu |\dot{\theta}_\mu|$ (larger time step). Growth continues until $L^2 < L^2_{\mathrm{cut}}$. We denote this rule by $\nu^\star = \mathrm{AVQDS}(\mathcal{C})$ for a candidate set $\mathcal{C}$. None of these elements — the equations of motion, the regulariser, the step control, the threshold, the trial arithmetic, or the pool — is modified in this work. The only quantities the new method reads are the trial distances L^2_ν that the loop computes anyway.

2.2 Problem statement and structural limits

A device is a connected coupling graph $G = (Q, E)$ with $|Q| = n$ and graph distance d_G. A *layout* π is a bijection from logical qubits (spin sites) to physical qubits. A compiled circuit $C = (c_1, \ldots, c_m)$ is a sequence of physical gates, each either a rotation $e^{-i\theta_\mu \hat{A}_\mu}$ placed on $\pi_k(\mathrm{supp}\hat{A}_\mu)$, with π_k the layout in force before position k, or a SWAP on an edge of E (which exchanges the logical content of its two qubits and updates the layout). C is *compliant* if every two-qubit gate acts on an edge of E. Costs follow the convention of the original method: a two-qubit Pauli rotation is two CNOTs plus single-qubit gates, a SWAP is three CNOTs [33, 34], and depth is the as-soon-as-possible depth of the resulting {CNOT, U(2)} circuit with consecutive single-qubit gates on a qubit merged. The goal is a compliant circuit that prepares the same family of states as AVQDS to within the AVQDS accuracy control, at the lowest CNOT count and depth.

The two-qubit generators of the Hamiltonian pools sit on the bonds of the interaction graph: an 8-cycle for the periodic mixed-field Ising model (MFIM) and an 8-vertex path for the open Lieb–Schultz–Mattis (LSM) chain. Whether a static layout can make every bond a device edge is a Hamiltonian-cycle / Hamiltonian-path question [43]. The 2×4 lattice (topology A, Fig. 1) contains the Hamiltonian cycle $0{-}1{-}2{-}3{-}7{-}6{-}5{-}4$, so both interaction graphs embed with all bonds adjacent. The heavy-hex subgraph (topology B) is a tree, and the following elementary bounds hold for every static layout:

Lemma 1 (static embeddings into a tree)**.** *Let T be a tree on n vertices with diameter D and let $\lambda(T)$ be the largest number of edges of a linear forest (a subgraph of maximum degree two without cycles) contained in T. For every layout, (i) at most $\lambda(T)$ bonds of an n-cycle or of an n-path are device edges; (ii) the total excess distance satisfies $\sum_{\mathrm{bonds}}(d_G - 1) \geq n - 2$ for the cycle and $\geq n - 1 - D$ for the path. For topology B, $\lambda = 5$ and $D = 5$: at least three ring bonds and two chain bonds are non-adjacent, with total excess at least 6 and 2, respectively.*

Proof. The adjacent bonds form a subgraph of T of maximum degree two without cycles, which proves (i). A closed walk through all vertices of a tree crosses every edge at least twice, so the cycle's bond distances sum to at least $2(n-1)$; an open walk crosses every edge at least twice except those of one path, whose length is at most D, so the path's distances sum to at least $2(n-1) - D$. For B, every linear forest must drop one of the three edges at each of the two degree-three vertices 1 and 5, and these vertices are not adjacent, so $\lambda = 7 - 2 = 5$; exhaustive enumeration of all 8! layouts confirms the bounds are attained (576 ring and 216 chain layouts reach coverage five). □

Lemma 1 says that on B any compliant circuit that keeps using all bonds must move qubits during the simulation; the open question is how much that costs and who decides. Post-hoc routing takes the AVQDS circuit as given. The method below instead lets the ansatz construction take part in the decision.

2.3 Near-degeneracy of the trial step

The handle we exploit is a property of the trial-append loop itself. Let L^2_0 be the distance before a decision and $g_\nu = L^2_0 - L^2_\nu$ the gain of candidate ν. Two kinds of freedom are present at almost every decision:

- *Near-degenerate gains.* Symmetry-related operators (bonds or sites related by translation in the periodic MFIM, or by the approximate symmetry of the bulk of the LSM chain) have equal or nearly equal gains, and generators whose tangent vectors $\hat{A}_\nu|\Psi\rangle$ have similar overlaps with the McLachlan residual have similar gains. The original authors already noted that the list $\{L^2_\nu\}$ "often contains almost degenerate minimal values" [21].

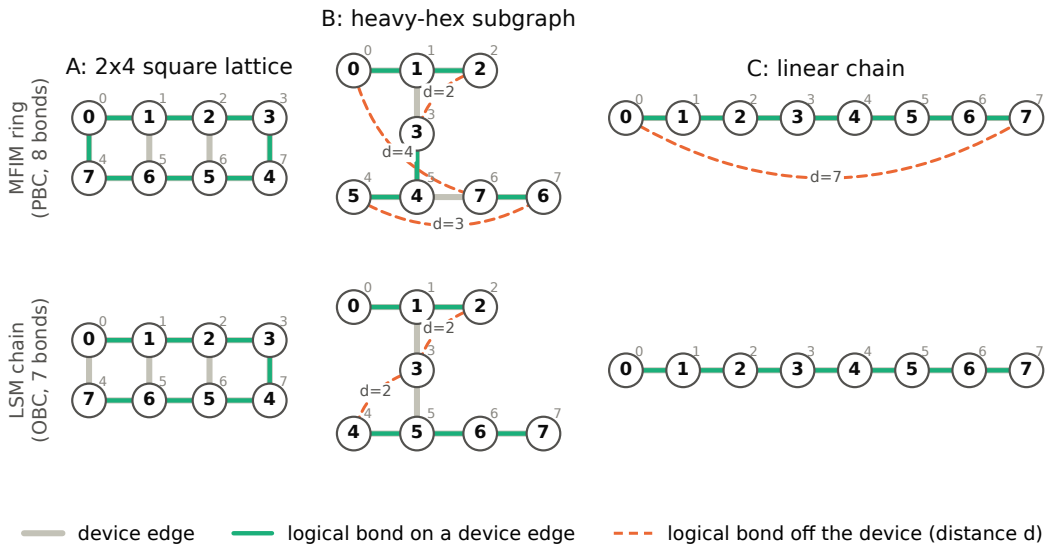


Figure 1: Coupling graphs and the a-priori layouts used by CA-AVQDS. Node labels are logical sites; small grey numbers are physical qubits. Green: logical bonds that are device edges. Dashed orange: bonds that are not, with their graph distance d. On A every bond is an edge. On B (Lemma 1) three ring bonds and two chain bonds are not. On C the ring's periodic bond has $d = 7$.

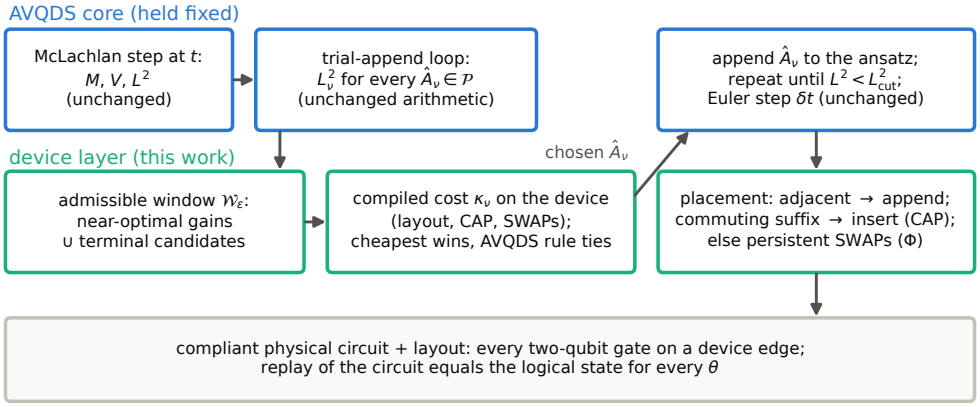


Figure 2: One growth decision of CA-AVQDS. The AVQDS core (blue) is executed unchanged; the device layer (green) only chooses among admissible candidates and compiles the chosen rotation onto the device.

- *Terminal decisions.* Growth is triggered as soon as L^2 creeps above L^2_{cut}, so at the decisive moment many candidates bring L^2 back below the threshold. The original rule already lets such a candidate displace the largest-gain operator when it permits a larger time step (smaller $\max|\dot{\theta}|$), so it treats these candidates as acceptable.

Which member of a degenerate or terminal set is appended is therefore decided by a tie-break that carries no information about accuracy — and, as we show in Sec. 5.4, by floating-point round-off. CA-AVQDS spends this freedom on hardware cost.

2.4 The CA-AVQDS growth policy

CA-AVQDS maintains, next to the logical ansatz (1), a compliant physical circuit C and the current layout π (the layout after the last gate of C). Its logic is confined to two hooks of the trial-append loop (Fig. 2). *Selection* is called once per decision, after the unchanged trial arithmetic has produced $\{L^2_\nu\}$; it returns the operator to append. *Placement* is called after the operator has been appended to the logical ansatz and compiles it into C.

Compiled cost. For candidate ν let $s_\nu(C) \in \mathbb{N}_0$ be the number of SWAPs needed to execute $e^{-i\theta\hat{A}_\nu}$ compliantly given C: $s_\nu = 0$ if $\hat{A}_\nu$ is single-qubit, if its two qubits are adjacent under π, or if it can be placed by commutation (below); otherwise $s_\nu = d_G(\pi(i), \pi(j)) - 1$ for $\mathrm{supp}\hat{A}_\nu = \{i, j\}$. Two cost models are considered:

$$\kappa_\nu^{\mathrm{swap}} = s_\nu, \qquad \kappa_\nu^{\mathrm{cnot}} = 3s_\nu + 2\,\big[\,|\mathrm{supp}\,\hat{A}_\nu| = 2\,\big], \tag{3}$$

where $[\cdot]$ is the Iverson bracket (1 if the statement is true, 0 otherwise). κ^{swap} only penalises routing; κ^{cnot} is the exact increment of N_{cx} caused by the decision, including the two CNOTs of the rotation itself.

Admissible window. With $g_{\max} = \max_\nu g_\nu$ and a strict-gain margin $\eta = 10^{-8}$ (the improvement margin of the original code),

$$\mathcal{W}_\epsilon = \big\{\nu :\ g_\nu > \eta \ \text{ and } \ \big[g_\nu \geq (1-\epsilon)\, g_{\max} \ \text{ or } \ (\tau \text{ and } L_\nu^2 < L_{\mathrm{cut}}^2)\big]\big\}, \tag{4}$$

where $\epsilon \in [0, 1]$ is the relative width of the near-degeneracy window and $\tau \in \{0, 1\}$ switches the admission of terminal candidates.

Selection rule. Let $\nu^\star = \textsc{avqds}(\mathcal{P})$ be the operator the original method would append and $\kappa_{\min} = \min_{\nu\in\mathcal{W}_\epsilon} \kappa_\nu$. CA-AVQDS appends

$$\nu_{\mathrm{CA}} = \begin{cases} \nu^\star & \text{if } \kappa_{\nu^\star} = \kappa_{\min}, \\ \textsc{avqds}\big(\{\nu \in \mathcal{W}_\epsilon : \kappa_\nu = \kappa_{\min}\}\big) & \text{otherwise.} \end{cases} \tag{5}$$

The rule is lexicographic: accuracy first (only admissible operators compete), compiled cost second, the original rule third. It deviates from AVQDS only when an admissible operator is strictly cheaper, and never trades accuracy against cost through a penalty weight. The loop still stops only when $L^2 < L_{\mathrm{cut}}^2$.

Commutation-aware placement (CAP). A two-qubit rotation that is not adjacent under the final layout may still be executable without SWAPs earlier in the circuit. Scanning C backwards from its end, the scan stops at the first rotation whose generator does not commute with $\hat{A}_\nu$ (two Pauli strings commute iff they anticommute on an even number of sites); SWAPs are traversed by undoing them on the layout. The rotation is inserted at the latest position p reached by the scan at which $\pi_p(i)$ and $\pi_p(j)$ are adjacent. Since $[\hat{A}_\nu, \hat{A}_\mu] = 0$ implies $[e^{-i\theta\hat{A}_\nu}, e^{-i\phi\hat{A}_\mu}] = 0$ for all angles, the compiled unitary equals that of the logical ansatz for *every* parameter vector. The derivative states, M, V and therefore the dynamics are unchanged. The logical ansatz keeps the operator at its end. CAP is optional; the development-set selection (Sec. 2.6) switched it off, and we analyse why in Sec. 5.1.

Persistent routing with McLachlan-weighted look-ahead. If neither adjacency nor CAP applies, SWAPs are inserted at the end of C along a shortest path and the layout is kept afterwards (no swap-back). All options are enumerated: every shortest path P from $\pi(i)$ to $\pi(j)$ and every split $m \in \{0, \ldots, d-1\}$ in which i moves m steps along P and j moves $d-1-m$ steps towards it. The option is chosen by minimising the potential of the resulting layout π',

$$\Phi(\pi') = \sum_{\nu\in\mathcal{P}_2} w_\nu \,\big[d_G\big(\pi'(i_\nu), \pi'(j_\nu)\big) - 1\big], \qquad w_\nu = 10^{-3} + \max(g_\nu, 0), \tag{6}$$

where $\mathcal{P}_2$ is the set of two-qubit pool operators and the weights are the McLachlan gains of the current decision ("trial demand"): bonds the dynamics currently asks for are kept close. A uniform-weight variant ($w_\nu = 1$) and the naive choice (first BFS path, first qubit moves) are ablations.

Initial layout. The initial layout is fixed a priori, without looking at any circuit: maximise the number of Hamiltonian-pool bonds that are device edges, then minimise $\sum(d_G - 1)$, then $\sum(d_G - 1)^2$ (penalising long bonds), then take the lexicographically smallest permutation. On A this is the Hamiltonian-cycle (MFIM) or Hamiltonian-path (LSM) embedding; on B it attains the bounds of Lemma 1 (Fig. 1). The quadratic tie-break was added after we inspected one early CA-AVQDS run on a headline configuration (documented in our attempts log); Sec. 5.4 reports the sensitivity to the initial layout.

2.5 Properties

Proposition 1. *(a)* Compliance: *every rotation is placed on adjacent qubits and every SWAP on an edge, so C is compliant by construction. (b)* Accuracy control: *the sequence of states satisfies the same per-step criterion as AVQDS, $L^2 < L_{\mathrm{cut}}^2$ after growth, with the same equations of motion; only the choice among admissible generators differs. (c)* Exact recovery: *if every two-qubit pool operator is adjacent under π_0 (e.g. topology A with the cycle/path embedding) and $\kappa = \kappa^{\mathrm{swap}}$, all costs vanish, no deviation, routing or CAP occurs, and CA-AVQDS reproduces AVQDS operation by operation. With κ^{cnot} on such an embedding it reproduces CNOT-aware AVQDS on the complete graph. (d)* Equivalence of the compiled circuit: *with the final layout used to relabel measurement outcomes, C prepares exactly the logical state $|\Psi[\boldsymbol{\theta}]\rangle$ for all $\boldsymbol{\theta}$. (e)* Classical overhead: *per decision the policy costs $O(|\mathcal{P}|\,|C|)$ for the CAP scans and $O(|\mathcal{P}_2|\,|\mathcal{O}|)$ for the look-ahead ($|\mathcal{O}|$ routing options), against $O(|\mathcal{P}|\,N_\theta^3)$ for the pseudo-inverses of the trial loop.*

(a)–(c) follow from the construction and (d) from the CAP argument above plus the fact that SWAPs only relabel qubits. We verify (d) numerically for every reported run by executing C on a statevector simulator and comparing it with the logical state, and we verify (c) for the implementation by bit-identical replay (Sec. 3).

2.6 Hyper-parameters and how they were fixed

The policy has five discrete switches and one continuous parameter: selection on or off, ϵ, τ, the cost model, CAP on or off, and the look-ahead (Φ with trial or uniform weights, or none). They were fixed *before* any headline run, on a development set that shares no configuration with the results reported later: $N = 6$ MFIM ($h_z = 0, 0.5$) and LSM ($h_z = -0.7, 1.6$) on a 6-qubit heavy-hex fragment (a tree with one degree-three vertex) and on a 2×3 lattice. Stage 1 screened 150 designs (one pool order each; $\epsilon \in \{0, 0.1, 0.3, 0.5, 0.7, 1\}$). Stage 2 re-evaluated the best 10 designs and the best design of each structural family with five pool orders each. The pre-registered rule was: discard designs that fail to complete within budget, violate compliance, or lose more than 0.005 final fidelity relative to AVQDS on any development run; then minimise the mean CNOT multiplier; break ties by depth and simplicity. The selected design is $\epsilon = 0.7$, $\tau = 1$ (terminal candidates admitted), $\kappa = \kappa^{\mathrm{cnot}}$, CAP off, and look-ahead with the trial-weighted Φ of Eq. (6). All headline runs use exactly these values together with the AVQDS hyper-parameters of Sec. 2.1. Appendix B lists both stages.

2.7 Relation to prior work

Post-hoc mapping and routing. Layout and routing heuristics and exact solvers — SABRE [35, 44], t|ket⟩ [36, 45], A* and SAT/SMT formulations [37, 38, 46], token swapping [39, 47] and noise-aware mapping [40, 41] — transform a *fixed* circuit. They cannot change which logical gates appear, and this is where the largest savings in our experiments come from. We therefore use them as references: the shortest-path baseline, and SABRE applied both to the unconstrained AVQDS circuit and to the CA-AVQDS circuit. *Structure-aware compilers* for Hamiltonian simulation and QAOA exploit the commutation of Trotter terms to reorder gates and co-optimise routing: 2QAN [48], Paulihedral [49], Tetris [50], term grouping [51], phase-gadget synthesis [52], commutation-aware mapping [53] and swap networks [54–59]. CAP is a deliberately minimal, online member of this family: only the newly grown gate moves, and only within its commuting suffix. It is exact for every parameter vector, so it is compatible with variational equations of motion. In our benchmarks it helps a pure router, but next to the selection rule its effect is within the scatter across tie-breaking orders (Sec. 5.1). *Hardware-aware ansätze.* Hardware-efficient ansätze [60] avoid routing by construction, and adaptive methods with hardware-efficient pools (qubit-ADAPT [27]) reduce circuit cost. Restricting a Hamiltonian pool to device edges, however, sacrifices the Hamiltonian bias that makes AVQDS compact and cannot represent the off-device bonds of Lemma 1 at all. In our tests the AVQDS loop then aborts early (Sec. 5.1). Closest to our selection rule is the hardware-aware qubit-ADAPT-VQE of Weaving et al. [61], which multiplies the energy gradient by $(1 - s/W)^b$, with s/W an approximate subgraph-isomorphism penalty, and transpiles afterwards. That work concerns a static ground-state cost function and uses a soft penalty with a strength b, which alters selections even when costs tie. CA-AVQDS differs in three ways. It acts on a time-dependent cost (McLachlan's distance) and keeps the accuracy criterion as a hard constraint. It tracks the device layout online, so that costs, CAP and routing reflect the actual compiled circuit. And it reduces exactly to the original algorithm when costs tie

(Proposition 1c). Related co-design ideas appear in [62, 63]. TETRIS-type layering [25, 64], adaptive pVQD [15] and other adaptive dynamics [22–24] do not address connectivity.

3 Experimental setup

Benchmarks. We use the two benchmark families of the original AVQDS study [21] without modification. (i) The mixed-field Ising ring $\hat{H} = -J\sum_i \hat{Z}_i\hat{Z}_{i+1} + h_x\sum_i \hat{X}_i + h_z\sum_i \hat{Z}_i$ with $N = 8$, periodic boundaries, $J = 1$, $h_x = -2$ and $h_z \in \{0, 0.5\}$ (integrable transverse-field limit [65] and nonintegrable case), quenched at $t = 0$ from $|0\rangle^{\otimes 8}$ and evolved to $T = 3$ [66]. The pool is the Hamiltonian pool: $\{\hat{Z}_i\hat{Z}_{i+1}\}$ (8 bonds incl. $\hat{Z}_7\hat{Z}_0$) and $\{\hat{X}_i\}$ (16 operators), plus $\{\hat{Z}_i\}$ for $h_z = 0.5$ (24 operators). (ii) The Lieb–Schultz–Mattis chain [67, 68] $\hat{H} = -J\sum_{i=0}^{N-2}[(1+\gamma)\hat{X}_i\hat{X}_{i+1} + (1-\gamma)\hat{Y}_i\hat{Y}_{i+1}] + h_z\sum_i \hat{Z}_i$ with $N = 8$, open boundaries, $J = 1$, $h_z \in \{-0.7, 1.6\}$, a linear ramp $\gamma(t) = 1 - 2t/T$ for $t \le T = 3$ followed by evolution under $\hat{H}(T)$ for another T (final time 6). The initial state is the exact ground state of $\hat{H}(0)$. The pool is $\{\hat{Z}_i\} \cup \{\hat{X}_i\hat{X}_{i+1}, \hat{Y}_i\hat{Y}_{i+1}\}$ (22 operators). Hamiltonians, pools and initial states are generated by the authors' code.

Topologies. Topology A is the 2×4 square lattice (10 edges), topology B the 8-qubit heavy-hex subgraph with edges $\{01, 12, 13, 35, 45, 56, 67\}$ (a tree; Lemma 1), and, as an out-of-sample third topology never used during development, C is the linear chain $0-1-\cdots-7$. Spin sites are logical qubits; mapping them onto a topology is part of each method. Figure 1 shows the a-priori layouts of CA-AVQDS. The ring embeds perfectly on A (16 optimal layouts), with coverage $5/8$ on B (576 layouts reach it) and $7/8$ on C (the periodic bond at distance 7). The chain embeds perfectly on A (28) and C (2), with coverage $5/7$ on B (216).

Methods compared. *Unconstrained AVQDS* is the authors' algorithm and hyper-parameters, counted with the all-to-all convention (2 CNOTs per two-qubit rotation). *Smart-embedding naive routing* takes the unchanged unconstrained circuit and, for every maximum-coverage embedding, inserts SWAPs along a shortest path in front of every non-adjacent two-qubit rotation. Four standard insertion variants are tried: swap-in/swap-out, and persistent movement of the first, second, or both qubits. We report the best combination per circuit, the strongest member of this baseline class; the full distribution is logged. *SABRE-routed AVQDS* is an additional reference that is much stronger than the required baseline: Qiskit's SABRE layout and routing [35, 44, 69] (200 layout and 200 swap trials, routing only, no gate optimisation) applied to the same unconstrained circuit. *CA-AVQDS* is the method of Sec. 2 with the hyper-parameters fixed on the development set. Exact reference states come from exact diagonalisation (MFIM) and a fourth-order Magnus integrator [70, 71] with steps $\le 10^{-3}$ (LSM), converged to 10^{-13} in fidelity against steps of 5×10^{-4}.

Pairing and seeds. AVQDS is deterministic, but its output depends on how ties are broken. We therefore run every method with five pool orders: seed 0 is the authors' order, and seeds 1–4 are fixed random permutations of the same pool, which change only the tie-breaking order. Within a seed, the unconstrained run, its naive and SABRE routings and the CA-AVQDS runs share the initial state, the pool order and the time-stepping rule ($\delta\theta_{\max}$, $\delta t_{\max}$, Euler). The naive and SABRE circuits *are* the unconstrained circuit, so they also share its time grid. Headline tables show seed 0; overhead multipliers are reported as mean and standard deviation over all five seeds, each relative to the unconstrained run of the same seed.

Metrics and audits. Per time step we log the fidelity $f(t) = |\langle\Psi_{\text{exact}}(t)|\Psi[\boldsymbol{\theta}(t)]\rangle|^2$, $N_{\text{cx}}(t)$, depth, N_θ, the Loschmidt echo $|\langle\Psi_0|\Psi(t)\rangle|^2$ (MFIM), the energy and spin correlations $\langle\hat{X}_0\hat{X}_1\rangle, \langle\hat{X}_0\hat{X}_7\rangle, \langle\hat{Y}_0\hat{Y}_1\rangle, \langle\hat{Y}_0\hat{Y}_7\rangle$ (LSM). Fidelities are recorded only; no decision of any method reads them. For every compiled circuit we store each two-qubit gate with the step at which it was added, its logical and physical qubits and an edge-set flag (compliance audit). We then replay the complete compiled circuit on a statevector and compare it with the logical variational state (replay overlap).

Implementation and verification. The authors' implementation is reused unchanged for the equations of motion, regularisation, step control, trial arithmetic, pools and models. Only the dense

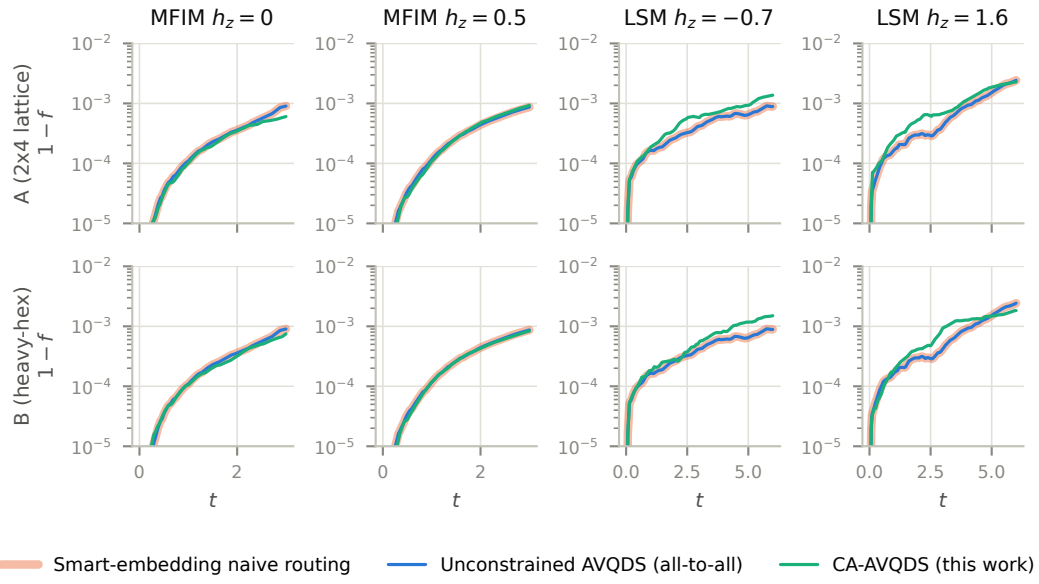


Figure 3: Infidelity $1 - f(t)$ against the exact state for the four benchmarks on topologies A (top) and B (bottom), seed 0. The naive and SABRE routings execute the unconstrained circuit, so their curves coincide with it (wide orange line underneath).

operator container was replaced by an exact Pauli-string container ($e^{\alpha \hat{P}} = \cosh \alpha + \sinh \alpha \, \hat{P}$), a speed-up of more than two orders of magnitude (140–230×). On the same backend, our refactored trial loop with the policy switched off is bit-identical to the authors' class. The fast backend reproduces the original backend's operator sequences on three benchmarks; on MFIM $h_z = 0.5$ it splits after 18 operators at a round-off tie (Sec. 5.4). The policy is a separate object attached to the trial loop, and a single switch (`policy=None`) recovers AVQDS. All runs are noiseless statevector simulations on CPUs, single-threaded per run.

4 Experimental results

Accuracy is preserved on every run. Across all 60 CA-AVQDS runs (four benchmarks, three topologies, five pool orders), the final-time fidelity differs from that of unconstrained AVQDS with the same pool order by between -0.0014 and $+0.0006$ (mean -0.0002; it is lower in 32 runs and higher or equal in the others). The worst case is thus seven times smaller than the 1% tolerance that would still count as accuracy-preserving. The infidelity curves in Fig. 3 track those of AVQDS over the whole trajectory, not only at the final time: the McLachlan threshold controls the error step by step for both methods. The physical observables follow the exact dynamics equally well (Figs. 6 and 7). The largest deviation of the Loschmidt echo from the exact echo is 5.8×10^{-3} for CA-AVQDS (all seeds and topologies), against 6.9×10^{-3} for AVQDS. The LSM energy and spin correlations, including the slowly built-up end-to-end correlation $\langle \hat{Y}_0 \hat{Y}_7 \rangle$, are reproduced through the ramp and the post-ramp evolution. The lowest MFIM fidelity reached anywhere along any CA-AVQDS trajectory is 0.9991.

Every compiled circuit is compliant. The audits cover every two-qubit gate of every compiled circuit: 12,136 gates in 220 circuits on A, 27,093 in 316 on B and 17,837 in 220 on C, including naive, SABRE, ablation and embedding runs. All gates lie on device edges (0 violations). Replaying each CA-AVQDS circuit on a statevector reproduces the logical variational state with overlap 1 to within 10^{-12} (Proposition 1d). Figure 8 shows where the two-qubit gates land. On A, CA-AVQDS uses only the edges of the Hamiltonian cycle (MFIM) or path (LSM). On B, the canonical runs need 59 SWAPs for MFIM $h_z = 0.5$ (naive: 108) and 5 for LSM $h_z = 1.6$ (naive: 26).

CNOT counts and depth. Tables 1 and 2 report the canonical runs (authors' pool order); Table 3 reports multipliers over all five pool orders. *Topology A.* The naive and SABRE baselines reproduce

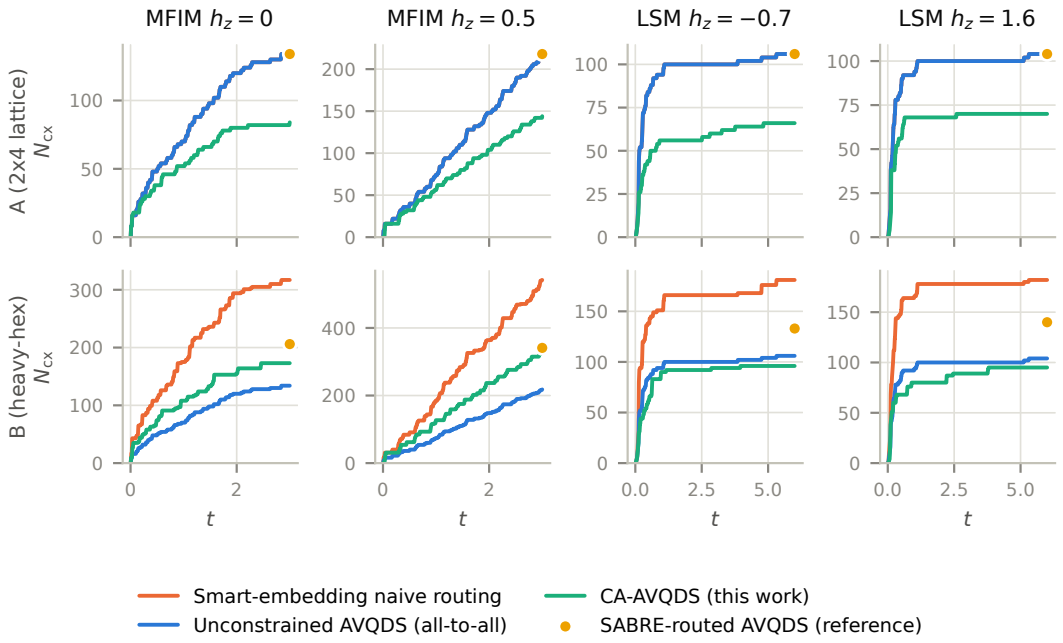


Figure 4: $N_{cx}(t)$ on A (top) and B (bottom), seed 0. On A the naive curve is identical to the unconstrained one. Yellow dots: SABRE routing of the final unconstrained circuit (SABRE has no time-resolved counterpart because it routes the complete circuit).

Table 1: MFIM ($N = 8$, PBC, $T = 3$): final-time fidelity, CNOT count N_{cx} and circuit depth (canonical pool order, seed 0). Naive routing executes the unconstrained circuit, so its fidelity equals the unconstrained one. The SABRE rows are an additional reference router applied to the same unconstrained circuit; the last row routes the logical CA-AVQDS circuit with SABRE instead of the online router (a post-hoc composite, see Sec. 5.1).

Method	Topology	MFIM $h_z = 0$			MFIM $h_z = 0.5$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.9991	134	85	0.9991	218	146
Smart-embedding naive routing	A	0.9991	134	85	0.9991	218	146
Smart-embedding naive routing	B	0.9991	317	216	0.9991	542	371
CA-AVQDS (this work)	A	0.9994	84	68	0.9991	144	100
CA-AVQDS (this work)	B	0.9993	173	123	0.9992	333	250
SABRE-routed AVQDS (ref.)	A	0.9991	134	85	0.9991	218	146
SABRE-routed AVQDS (ref.)	B	0.9991	206	158	0.9991	341	273
CA-AVQDS circuit + SABRE (post hoc)	B	0.9993	152	118	0.9992	267	214

the unconstrained counts exactly (multiplier 1.00), because the cycle/path embedding makes every bond an edge. CA-AVQDS needs *fewer* CNOTs than unconstrained AVQDS (mean multiplier 0.73, depth 0.75). For instance, MFIM $h_z = 0$ uses 84 instead of 134 CNOTs, and LSM $h_z = -0.7$ uses 66 instead of 106. Since nothing is routed on A, this gain is entirely due to the selection rule, which replaces two-qubit by admissible single-qubit rotations (Sec. 5.1). *Topology B.* Routing is unavoidable (Lemma 1). The strongest naive baseline doubles the CNOT count (mean 2.04, depth 2.10), and SABRE reduces this to 1.42 (depth 1.71). CA-AVQDS reaches 1.32 (depth 1.42). It uses fewer CNOTs than the naive baseline in 20 of 20 paired runs, by 35% on average (one-sided Wilcoxon signed-rank test $p = 4.4 \times 10^{-5}$), and fewer than SABRE in 13 of 20 runs (1 tie; $p = 2.3 \times 10^{-2}$). Against SABRE the advantage is clear for LSM (1.01 vs. 1.20 and 1.10 vs. 1.29) but absent for the MFIM ring (1.54 vs. 1.57 and 1.62 vs. 1.62), a case we return to in Sec. 5.2. The mean depth is lowest for CA-AVQDS on every benchmark of A and B. Figures 4 and 5 show that the savings in CNOTs and depth build up steadily during the growth phase, without abrupt events.

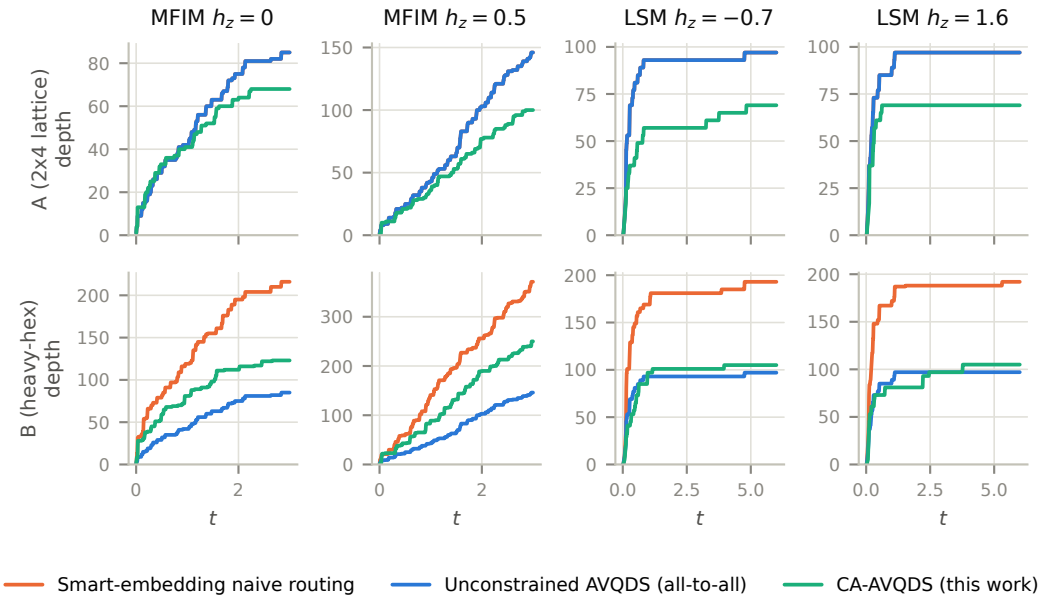


Figure 5: Circuit depth ({CNOT, U(2)} circuit, consecutive single-qubit gates merged) as a function of time on A (top) and B (bottom), seed 0.

Table 2: LSM ($N = 8$, OBC, ramp $T = 3$ plus post-ramp T, final time 6); same quantities and conventions as Table 1.

Method	Topology	LSM $h_z = -0.7$			LSM $h_z = 1.6$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.9991	106	97	0.9976	104	97
Smart-embedding naive routing	A	0.9991	106	97	0.9976	104	97
Smart-embedding naive routing	B	0.9991	181	193	0.9976	182	192
CA-AVQDS (this work)	A	0.9986	66	69	0.9977	70	69
CA-AVQDS (this work)	B	0.9985	96	105	0.9982	95	105
SABRE-routed AVQDS (ref.)	A	0.9991	106	97	0.9976	104	97
SABRE-routed AVQDS (ref.)	B	0.9991	133	149	0.9976	140	145
CA-AVQDS circuit + SABRE (post hoc)	B	0.9985	93	105	0.9982	92	101

Third topology. The linear chain C was never used during development (Table 4). For the LSM chain it admits a perfect embedding, and CA-AVQDS reproduces exactly its topology-A results (identical decisions, multipliers 0.71 and 0.70). The MFIM ring is the hard case: the periodic bond spans the whole device. The naive baseline needs 2.23 and 2.37 times the unconstrained CNOTs and SABRE 1.78 and 1.86, while CA-AVQDS needs 1.60 and 1.60. It beats the naive baseline in 20 and SABRE in 20 of the 20 runs on C ($p = 4.4 \times 10^{-5}$ for both). The method therefore carries over to an unseen topology without retuning.

5 Analysis

5.1 Mechanism: where the savings come from

An exact factorisation. For every pool order we also ran the selected policy on the complete graph, where no routing is needed. The CNOT multiplier of CA-AVQDS then factorises exactly as $m_{CA} = m_{sub}\, m_{topo}$. Here m_{sub} (complete graph vs. AVQDS) measures what the selection rule saves without any connectivity constraint, and m_{topo} (topology vs. complete graph) is the overhead the topology imposes on CA-AVQDS (Table 5). *Topology A* illustrates Proposition 1(c): in 20 of 20 runs the logical circuit on A is identical, operator by operator and parameter by parameter, to the complete-graph run, so $m_{topo} = 1$ and the whole gain on A, $m_{sub} = 0.73$, is topology-independent.

Table 3: Overhead multipliers relative to unconstrained AVQDS (all-to-all convention, 2 CNOTs per two-qubit rotation) of the same pool order. Mean ± standard deviation over the five pool orders (seed 0 is the authors' order; its values follow from Tables 1 and 2). Naive: best smart embedding and shortest-path insertion; SABRE: reference router applied to the same unconstrained circuits.

		CNOT multiplier			depth multiplier		
Benchmark	Top.	CA-AVQDS	naive	SABRE	CA-AVQDS	naive	SABRE
MFIM $h_z=0$	A	0.75 ± 0.09	1.00 ± 0.00	1.00 ± 0.00	0.83 ± 0.12	1.00 ± 0.00	1.00 ± 0.00
MFIM $h_z=0.5$	A	0.75 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	0.73 ± 0.05	1.00 ± 0.00	1.00 ± 0.00
LSM $h_z=-0.7$	A	0.71 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	0.72 ± 0.11	1.00 ± 0.00	1.00 ± 0.00
LSM $h_z=1.6$	A	0.70 ± 0.04	1.00 ± 0.00	1.00 ± 0.00	0.71 ± 0.17	1.00 ± 0.00	1.00 ± 0.00
mean	A	*0.73*	*1.00*	*1.00*	*0.75*	*1.00*	*1.00*
MFIM $h_z=0$	B	1.54 ± 0.22	2.41 ± 0.13	1.57 ± 0.05	1.76 ± 0.34	2.46 ± 0.30	1.97 ± 0.15
MFIM $h_z=0.5$	B	1.62 ± 0.15	2.49 ± 0.11	1.62 ± 0.04	1.68 ± 0.13	2.47 ± 0.13	1.93 ± 0.04
LSM $h_z=-0.7$	B	1.01 ± 0.12	1.63 ± 0.09	1.20 ± 0.04	1.07 ± 0.18	1.73 ± 0.21	1.43 ± 0.08
LSM $h_z=1.6$	B	1.10 ± 0.19	1.66 ± 0.11	1.29 ± 0.05	1.19 ± 0.15	1.74 ± 0.17	1.51 ± 0.12
mean	B	*1.32*	*2.04*	*1.42*	*1.42*	*2.10*	*1.71*

Table 4: Third topology (linear chain C, not used in any design decision). Seed-0 values; multipliers are mean ± sd over five pool orders.

Benchmark	fid. (CA)	N_{cx} unc.	naive	SABRE	CA-AVQDS	CX mult. CA-AVQDS	CX mult. naive
MFIM $h_z=0$	0.9993	134	290	233	207	1.60 ± 0.14	2.23 ± 0.15
MFIM $h_z=0.5$	0.9991	218	476	380	338	1.60 ± 0.09	2.37 ± 0.12
LSM $h_z=-0.7$	0.9986	106	106	106	66	0.71 ± 0.06	1.00 ± 0.00
LSM $h_z=1.6$	0.9977	104	104	104	70	0.70 ± 0.04	1.00 ± 0.00

Topology B carries a genuine routing overhead, $m_{\mathrm{topo}} = 1.80$ on average. This is smaller than the naive router's factor (2.04) but larger than SABRE's (1.42). Routing the leaner CA-AVQDS circuit costs relatively more, because the number of SWAPs is set by how often off-device bonds are needed, not by how many rotations the circuit contains. Our online router also routes less efficiently than SABRE (Sec. 5.2). The product still ends below SABRE's multiplier because the substitution factor more than compensates. On the out-of-sample chain C, $m_{\mathrm{topo}} = 1.57$ on average: 2.14 and 2.13 for the ring, and exactly 1 for the chain, which embeds perfectly.

What the selection rule actually does. Figure 9(a) classifies all growth decisions of the headline runs on B (3122 decisions). CA-AVQDS keeps the AVQDS choice in 34%. In 47% it replaces a two-qubit rotation by an admissible single-qubit rotation (on A: 50%). In 19% the AVQDS choice would have required SWAPs and an admissible adjacent two-qubit rotation is taken instead (on C: 9%; on A, by construction, never). The replacements are mostly not near-ties: the median ratio of the chosen to the largest gain is 0.20 (Fig. 9b). Most replacements in the selected design use terminal admissibility, i.e. an operator that already restores $L^2 < L^2_{\mathrm{cut}}$. The original rule already lets a terminal candidate displace the largest-gain operator if it comes later in the pool order and permits a larger time step; CA-AVQDS lets it do so when it is cheaper on the device. Terminal admission is nevertheless not indispensable. Switching it off raises the CNOT multiplier only from 0.73 to 0.82 on A and from 1.32 to 1.38 on B (Table 6), because the near-degenerate window then takes over most of the substitutions. Because a terminal operator with a small gain leaves L^2 just below the threshold, growth is triggered again sooner. The ansatz therefore ends with more parameters (N_θ ratio 1.24 on average) but fewer two-qubit rotations, and the per-step accuracy criterion is met throughout.

Component ablations. Table 6 and Fig. 10 isolate the switches; all numbers below are mean CNOT multipliers on B. (i) *Selection off.* The online router applied to the unchanged AVQDS circuit gives 1.93. This is close to the naive router (2.04) and far behind SABRE (1.42), so the router on its own is not competitive. Selection off also reproduces the AVQDS circuit operation by operation in 60 of 60 runs on all topologies, the second half of Proposition 1(c). (ii) *Look-ahead.* With selection on, the naive move (first path, first qubit) gives 1.73 and uniform potential weights give 1.44, compared with 1.32 for the selected trial-weighted potential. The two components are strongly synergistic. Selection with a myopic layout (1.73) and the look-ahead layout without selection (1.93) both remain far above

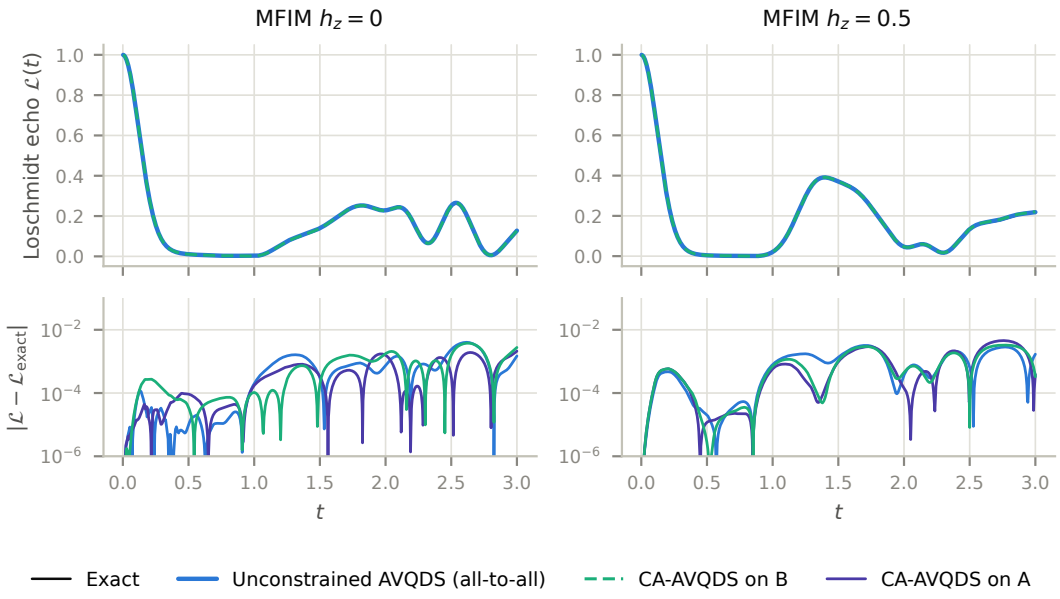


Figure 6: Loschmidt echo $\mathcal{L}(t) = |\langle\Psi_0|\Psi(t)\rangle|^2$ for the two MFIM quenches (seed 0; the exact curve is hidden under the variational ones) and its deviation from the exact echo.

Table 5: Where the CNOT reduction comes from (mean $\pm$ sd over five pool orders; all ratios use N_{cx}). $m_{\text{CA}} = N_{\text{cx}}^{\text{CA}}/N_{\text{cx}}^{\text{AVQDS}}$ factorises exactly into the substitution factor $m_{\text{sub}} = N_{\text{cx}}^{\text{CA, all-to-all}}/N_{\text{cx}}^{\text{AVQDS}}$ (the same policy on a complete graph, no routing) and the topology factor $m_{\text{topo}} = N_{\text{cx}}^{\text{CA}}/N_{\text{cx}}^{\text{CA, all-to-all}}$. 'SWAP-only' is CA-AVQDS with κ^{swap} (pure connectivity awareness), 'sel. off' keeps every AVQDS choice and only places/routes online.

Benchmark	Top.	m_{CA}	m_{sub}	m_{topo}	SWAP-only	sel. off	naive	SABRE
MFIM $h_z=0$	A	0.75 ± 0.09	0.75 ± 0.09	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
MFIM $h_z=0.5$	A	0.75 ± 0.06	0.75 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
LSM $h_z=-0.7$	A	0.71 ± 0.06	0.71 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
LSM $h_z=1.6$	A	0.70 ± 0.04	0.70 ± 0.04	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
MFIM $h_z=0$	B	1.54 ± 0.22	0.75 ± 0.09	2.05 ± 0.13	1.72 ± 0.12	2.46 ± 0.07	2.41 ± 0.13	1.57 ± 0.05
MFIM $h_z=0.5$	B	1.62 ± 0.15	0.75 ± 0.06	2.16 ± 0.14	1.69 ± 0.09	2.34 ± 0.05	2.49 ± 0.11	1.62 ± 0.04
LSM $h_z=-0.7$	B	1.01 ± 0.12	0.71 ± 0.06	1.42 ± 0.09	1.34 ± 0.06	1.49 ± 0.08	1.63 ± 0.09	1.20 ± 0.04
LSM $h_z=1.6$	B	1.10 ± 0.19	0.70 ± 0.04	1.56 ± 0.29	1.28 ± 0.07	1.44 ± 0.08	1.66 ± 0.11	1.29 ± 0.05
MFIM $h_z=0$	C	1.60 ± 0.14	0.75 ± 0.09	2.14 ± 0.19	1.71 ± 0.22	2.26 ± 0.11	2.23 ± 0.15	1.78 ± 0.05
MFIM $h_z=0.5$	C	1.60 ± 0.09	0.75 ± 0.06	2.13 ± 0.13	1.70 ± 0.14	2.34 ± 0.16	2.37 ± 0.12	1.86 ± 0.09
LSM $h_z=-0.7$	C	0.71 ± 0.06	0.71 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
LSM $h_z=1.6$	C	0.70 ± 0.04	0.70 ± 0.04	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

their combination (1.32). The trial-weighted potential keeps the bonds the dynamics currently asks for adjacent, so that later high-gain operators are free and the selection can take them without routing. Conversely, the selection avoids the operators the current layout makes expensive. The layout follows the dynamics and the ansatz follows the layout. (iii) *Terminal admission off.* The multiplier rises to 0.82 on A, 1.38 on B and 1.27 on C: terminal admission is useful but not essential. (iv) *SWAP-only cost.* Pure connectivity awareness, without single-qubit substitution, gives 1.51 on B and 1.35 on C. This is better than naive routing on both topologies and better than SABRE on C, but not on B. On A it reproduces AVQDS exactly (1.00). (v) *CAP.* Without selection, CAP places 4.6% of all operators earlier in the circuit and lowers the multiplier from 1.93 to 1.73. With selection on, it places only 1.4% and changes 1.32 to 1.29, within the scatter across pool orders. The selection rule already avoids most operators that CAP could rescue. In addition, the commuting suffix of a new two-qubit rotation is short: the single-qubit $\hat{X}_i$ (MFIM) and $\hat{Z}_i$ (LSM) rotations that fill these circuits anticommute with every bond that contains site i. On A, where every two-qubit pool operator is adjacent, the CAP

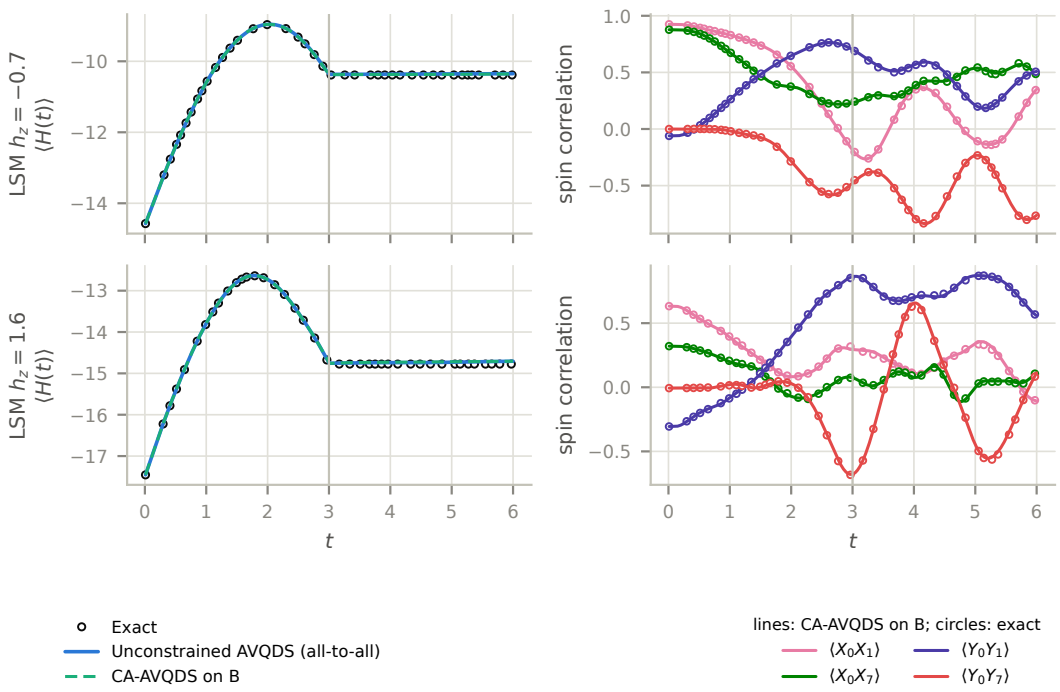


Figure 7: LSM ramp ($t \leq 3$) and post-ramp evolution: energy $\langle \hat{H}(t) \rangle$ and spin correlations of CA-AVQDS on topology B (lines) against exact results (circles), seed 0. The vertical line marks the end of the ramp.

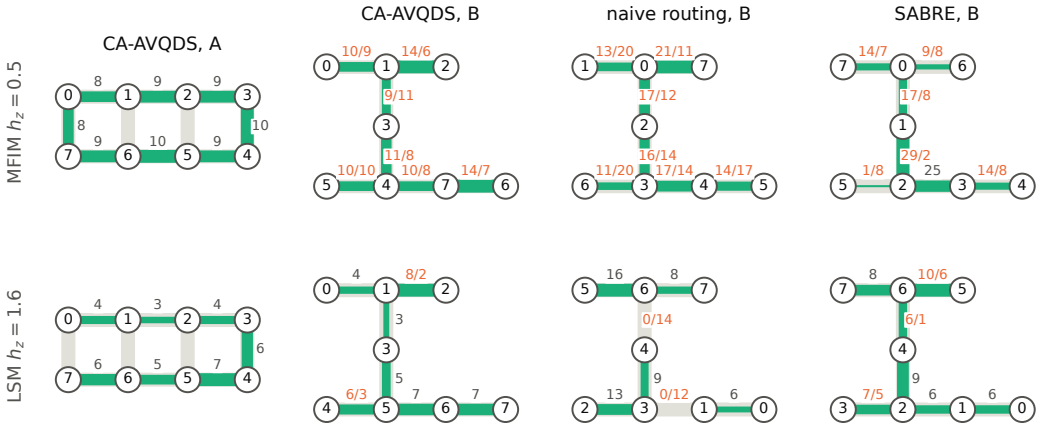


Figure 8: Where the two-qubit gates of the final circuits land (seed 0). Edge labels: number of two-qubit rotations / SWAPs on that edge; line width is proportional to the number of rotations; node labels: logical site initially on that qubit. On A, CA-AVQDS uses only the cycle edges (MFIM) or path edges (LSM). On B it needs far fewer SWAPs than naive routing.

and look-ahead switches cannot act. Running them anyway reproduces the corresponding designs operation by operation in all 80 of 80 runs.

Router independence. Is the advantage an artefact of comparing an ansatz-aware method with a fixed-circuit router? We routed the *logical* CA-AVQDS circuits with SABRE as well. The result is 1.18 on B and 1.09 on C, against 1.42 and 1.41 for the AVQDS circuits under the same router; the CA-AVQDS circuit is cheaper in 18 of 20 paired runs on B (one-sided Wilcoxon signed-rank $p = 1.1 \times 10^{-4}$) and 20 of 20 on C ($p = 4.4 \times 10^{-5}$). The logical circuits grown by CA-AVQDS are thus intrinsically cheaper to execute on the device, whichever router compiles them. The comparison

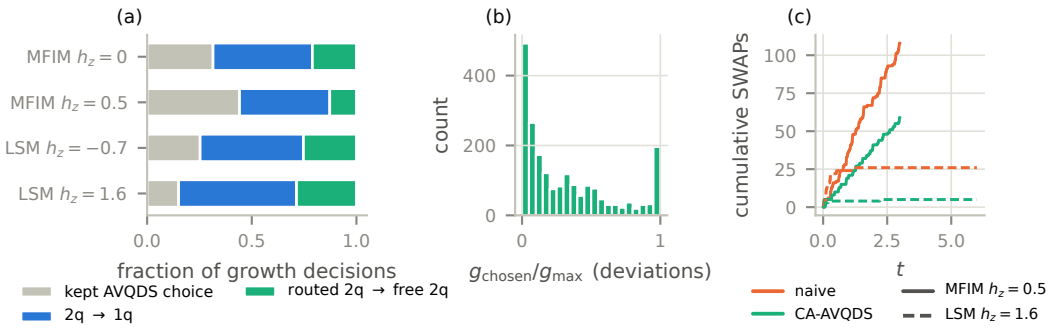


Figure 9: Mechanism on topology B (all five pool orders). (a) Fraction of growth decisions in which CA-AVQDS keeps the AVQDS choice, substitutes a single-qubit rotation for a two-qubit one, or takes an adjacent two-qubit rotation instead of one that needed SWAPs. (b) Ratio of the chosen to the largest McLachlan gain at the deviating decisions. (c) Cumulative SWAPs of CA-AVQDS and of naive routing, seed 0.

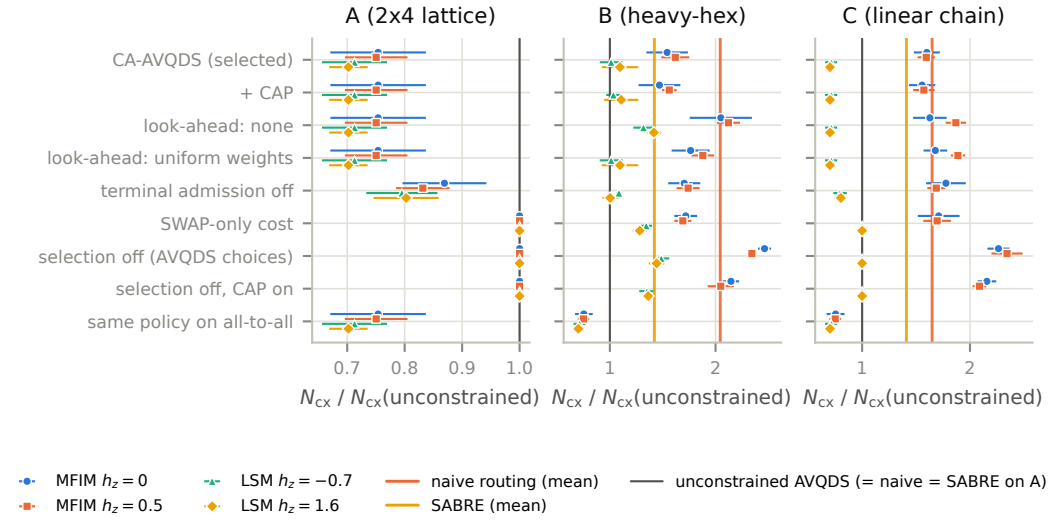


Figure 10: CNOT multiplier relative to unconstrained AVQDS for the selected design and each single-switch ablation (markers: benchmarks; bars: ± 1 sd over five pool orders). Vertical lines: mean naive and SABRE multipliers on that topology; 1.0 is unconstrained AVQDS.

also shows that SABRE routes them better than our online router, notably for the MFIM ring on B (Sec. 5.2).

Restricting the pool is not an option. The simplest way to obtain compliant AVQDS circuits is to drop every pool operator that is not a device edge under the static layout. We ran unconstrained AVQDS with such restricted pools on B and C. On every configuration with an off-device bond (all four benchmarks on B and both MFIM rings on C), the trial loop aborts within the first 0.09 time units with "dynamic ansatz cannot further improve": the remaining generators cannot represent the dynamics of the missing bonds, so L^2 cannot be brought below L^2_{cut}. It runs to completion only for the chain on C, where nothing is removed. Some form of routing is therefore indispensable, and the question is only how much of it the ansatz can avoid.

Why A and B behave differently. Topology A contains a Hamiltonian cycle, so routing never arises and only the topology-independent substitution acts. CA-AVQDS on A *is* CNOT-aware AVQDS. On the tree B, Lemma 1 guarantees that at every moment at least three ring bonds (two chain bonds) are off the device. The periodic MFIM quench is translation invariant, so every bond keeps being needed and routing cannot be avoided, only made cheaper. This is why m_{topo} stays well above one for the ring (2.05, 2.16) even though 13–21% of its decisions avoid a routed operator. The open LSM chain is structurally easier. Lemma 1 leaves only two of its bonds off the device, at distance

Table 6: Ablations: mean $\pm$ sd of the CNOT multiplier (relative to unconstrained AVQDS of the same pool order) over 4 benchmarks $\times$ 5 pool orders, worst final-fidelity change and mean parameter ratio $N_\theta / N_\theta^{\mathrm{AVQDS}}$ over the same runs. Each row changes one switch of the selected design. On A every two-qubit pool operator is adjacent, so the CAP and look-ahead switches cannot act there; they were run anyway and reproduce the corresponding designs exactly.

Variant	CX mult. A	CX mult. B	CX mult. C	$\Delta f_{\min}$	N_θ ratio
CA-AVQDS (selected design)	0.73 ± 0.07	1.32 ± 0.32	1.15 ± 0.46	-0.0014	1.24
selection off (AVQDS choices; online routing)	1.00 ± 0.00	1.93 ± 0.49	1.65 ± 0.68	$+0.0000$	1.00
selection off, CAP on	1.00 ± 0.00	1.73 ± 0.39	1.56 ± 0.58	$+0.0000$	1.00
CAP on	0.73 ± 0.07	1.29 ± 0.27	1.14 ± 0.45	-0.0014	1.25
look-ahead off (first path, first qubit)	0.73 ± 0.07	1.73 ± 0.41	1.23 ± 0.55	-0.0014	1.25
look-ahead with uniform weights	0.73 ± 0.07	1.44 ± 0.42	1.25 ± 0.56	-0.0014	1.25
terminal admission off ($\tau = 0$)	0.82 ± 0.07	1.38 ± 0.36	1.27 ± 0.49	-0.0007	1.06
SWAP-only cost κ^{swap}	1.00 ± 0.00	1.51 ± 0.22	1.35 ± 0.38	-0.0018	1.05
same policy on the complete graph	0.73 ± 0.07 (complete graph)			-0.0014	1.24

two (total excess 2, against 6 for the ring), so fewer routed operators are needed in the first place. Routing-driven choices (19% of all decisions on B over both models, more for LSM) then remove most of the remaining SWAPs: 4–6 per run on B against 16–26 for naive routing. As a result, m_{topo} is 1.42 and 1.56 for LSM. The same structural argument predicts the C results. The chain embeds perfectly ($m_{\mathrm{topo}} = 1$), while the ring's distance-7 periodic bond makes C about as costly as B for CA-AVQDS and more costly than B for SABRE.

5.2 Failure analysis

The online router is the weak link for the MFIM ring on B. For the two MFIM quenches on B, CA-AVQDS is not better than SABRE applied to the unconstrained circuit (1.54 vs. 1.57 and 1.62 vs. 1.62; it wins only 3 of 5 pool orders each). The router-independence test locates the problem. Routing the same CA-AVQDS circuits with SABRE gives 1.35 and 1.30, so the logical circuits are good but our persistent, one-step look-ahead router wastes SWAPs. It moves qubits for the decision at hand, whereas SABRE optimises the whole circuit with bidirectional passes and many trials. For the translation-invariant ring, whose bonds are all in constant demand, myopic moves keep undoing each other. A practical pipeline for such cases is CA-AVQDS for the ansatz followed by a strong router for execution (the post-hoc rows of Tables 1 and 2). We did not make this composite the headline method because it was not part of the pre-registered design.

Variance across pool orders. CA-AVQDS's multipliers on B scatter more across pool orders than the routers' (standard deviation up to 0.22 vs. 0.05 for SABRE; Table 3). The worst case is LSM $h_z = 1.6$ with pool order 4, where CA-AVQDS on B needs 133 CNOTs: 59 two-qubit rotations, against 34 on A for the same pool order. The routing-driven choices steered the trajectory into a region where more two-qubit generators were eventually required. Selection is greedy, and a locally cheap choice can raise the cost of later growth.

More parameters. The substitution trades two-qubit rotations for single-qubit ones and leaves L^2 closer to the threshold, so N_θ grows: by about $1.24\times$ on average, most for LSM (from about 70 to about 90–110 on average) and hardly for MFIM $h_z = 0.5$ (Table 10). Single-qubit rotations are cheap on hardware, but each parameter adds rows to M and V. The number of measurement circuits per step grows roughly as N_θ^2 [21, 72], and so does the classical cost (Sec. 5.3).

Fidelity. The fidelity changes are small but not all positive: 32 of 60 runs end slightly below AVQDS. For MFIM no run loses more than 10^{-4}. All larger losses occur for LSM, whose trajectory is twice as long, the worst being -0.0014 (LSM $h_z = -0.7$, pool order 1, on A and C). The per-step criterion is the same for both methods, but the more frequent growth events of CA-AVQDS, each ending just below L_{cut}^2, plausibly let more small errors accumulate.

CAP did not earn its place. Commutation-aware placement is exact and cheap, and it does help a pure router (item (v) above). Next to the selection rule, however, its effect is within the scatter across

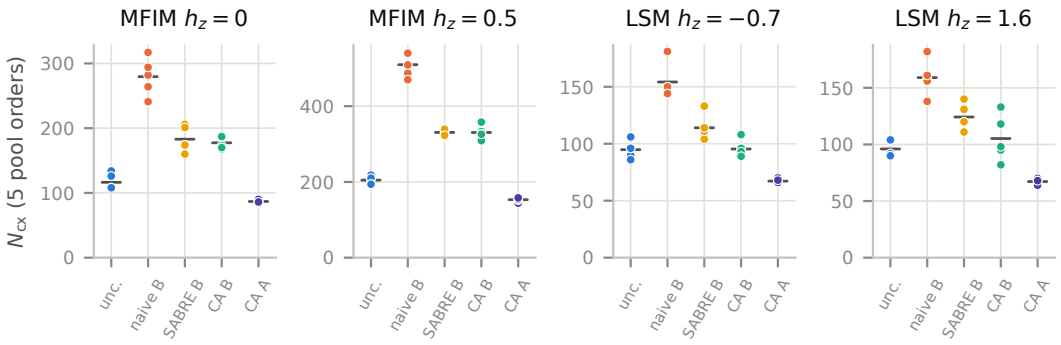


Figure 11: CNOT counts of all five pool orders (dots; bars: means) for unconstrained AVQDS, naive and SABRE routing on B, and CA-AVQDS on B and A.

pool orders, and the development set therefore did not select it. CAP should matter more in circuits with long commuting blocks, such as diagonal Ising-type or QAOA layers, which AVQDS with these Hamiltonian pools does not produce.

Scope of the substitution gain. The topology-independent part of the gain requires single-qubit generators in the pool that are admissible often enough. This holds for the Hamiltonian pools of spin models (an admissible single-qubit alternative existed at 76–91% of the AVQDS two-qubit choices). For pools without single-qubit generators, such as fermionic excitation pools, only the routing-driven part (SWAP-only cost, 1.51 on B) would remain.

5.3 Computational cost

The device layer itself is negligible. Summed over a run, its decision hooks (cost evaluation, window, look-ahead) take at most 0.03% of the wall-clock time, and its per-step bookkeeping of depth and audits takes well under a second. The total wall-clock time of CA-AVQDS relative to unconstrained AVQDS is nevertheless 1.41 (A), 1.40 (B) and 1.41 (C) on average, up to 1.80 for LSM. Since the number of time steps is essentially unchanged, the excess comes from the larger ansatz that the unchanged AVQDS core propagates. The derivative states cost $O(N_\theta^2)$ per step and the trial loop $O(|\mathcal{P}|N_\theta^3)$ per decision, so the 30–50% more parameters of LSM translate into about 1.6–$1.8\times$ the time, while MFIM (about 10% more parameters or fewer) costs 1.0–$1.3\times$. In absolute terms every run takes between half a minute and two minutes on a single core (78 s on average for the largest unconstrained benchmark). By comparison, SABRE with 200 layout and 200 swap trials routes one circuit in under a second. On the cost–quality plane CA-AVQDS therefore sits at slightly higher classical cost than AVQDS and SABRE, and at a substantially lower quantum cost (CNOTs and depth) than either baseline on every topology.

5.4 Robustness

Pool order. Permuting the pool changes only the tie-breaking order, yet it changes the unconstrained CNOT count substantially: MFIM $h_z = 0$ gives 134, 108, 106, 108, 126 CNOTs for the five orders, and the authors' order is the most expensive one (or tied for it) in all four benchmarks. All comparisons in this paper are therefore paired by pool order and aggregated over all five (Fig. 11). This sensitivity, together with our quench timing at $t = 0$ instead of $t = 0.01$ (Appendix A), is the likely reason why our MFIM $h_z = 0.5$ count (218) differs from the published 210 although the algorithm and hyper-parameters are the same.

Initial layout on B. The a-priori layout is one of 576 (ring) or 216 (chain) maximum-coverage layouts. We reran CA-AVQDS (pool order 0) from 24 randomly sampled maximum-coverage layouts per benchmark and compared it with the best naive routing from the same layouts (Fig. 12). CA-AVQDS stays below the naive router for all 96 of 96 layouts. Its multipliers range over 1.24–1.57, 1.36–1.71, 0.89–1.46 and 0.90–1.41 for the four benchmarks (naive: 2.48–3.04, 2.58–3.04, 1.74–2.44, 1.95–3.05), and the a-priori layout lies below the median in every case (ranks 4, 8, 1 and 1 of 24). Its quadratic-excess tie-break was adopted after we inspected one early CA-AVQDS run on a headline

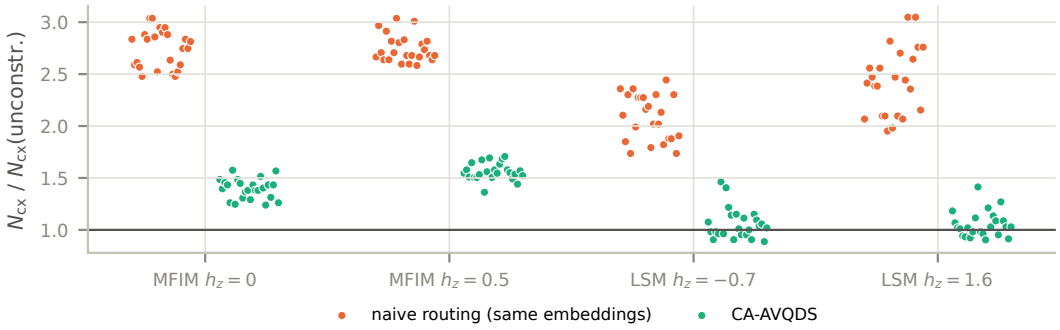


Figure 12: Sensitivity to the initial layout on B (pool order 0): CNOT multiplier of CA-AVQDS and of the best naive routing for the same 24 sampled maximum-coverage layouts per benchmark.

configuration, as documented in our attempts log. The unbiased reference is therefore the median over random maximum-coverage layouts: 1.40, 1.55, 1.01 and 1.02, against 2.78, 2.69, 2.12 and 2.43 for naive routing. The largest fidelity loss over these 96 runs is -0.0017, again far inside the 1% tolerance.

6 Conclusion

The decisions that grow an adaptive variational circuit are far less constrained than the greedy rule that makes them suggests. In AVQDS most growth decisions admit several operators that restore McLachlan's accuracy threshold, and the original rule picks among them by a tie-break that is ultimately decided by round-off. CA-AVQDS gives that choice to the device. Admissibility first, compiled CNOT cost on the device second, the original rule third, plus a persistent layout with a McLachlan-weighted look-ahead for the routing that remains. The McLachlan equations of motion and the accuracy criterion stay untouched, the circuits are compliant by construction, and the method collapses to AVQDS when costs tie.

On the original AVQDS benchmarks, the variant preserves accuracy everywhere (worst final-fidelity change -0.0014). On average over benchmarks and tie-breaking orders it needs fewer CNOTs and less depth than unconstrained AVQDS compiled by either the strongest shortest-path router or SABRE, on all three topologies. The gain has two clearly separated parts. A topology-independent substitution of two-qubit by admissible single-qubit rotations is the only effect on the square lattice, where a Hamiltonian cycle makes routing unnecessary and CA-AVQDS beats even the unconstrained algorithm. A topology-dependent part dominates where routing is unavoidable, as on the heavy-hex tree. The logical circuits CA-AVQDS grows stay cheaper when compiled by an independent state-of-the-art router, so the improvement is not an artefact of our routing. At the same time, that router is better than ours for the translation-invariant ring on heavy-hex, which is the clearest failure of the online design.

Limitations. All results are noiseless statevector simulations of eight-qubit systems. We count CNOTs and depth but ignore the cost of single-qubit gates, which CA-AVQDS uses more of, and the larger parameter count (N_θ about $1.5\times$ larger on average for LSM, up to $1.9\times$ in one run) increases measurement and classical cost. The selection is greedy and occasionally steers the trajectory into costlier regions (Sec. 5.2). The online router uses only a one-step look-ahead. The substitution gain presupposes single-qubit generators in the pool. Hyper-parameters were selected on six-qubit analogues of the benchmarks and transferred without retuning; the third topology suggests that this transfer is benign, but larger and noisier settings remain to be tested.

Outlook. Three extensions follow directly. (i) Replacing CNOT counts in κ by calibrated per-edge error rates turns the same rule into a noise-aware ansatz construction. (ii) Coupling the growth to a global router, for example by using SABRE's cost of the current circuit as the look-ahead potential, should close the gap identified for the ring. (iii) The rule only requires the trial distances of an adaptive method, so it applies unchanged to AVQDS(T), adaptive imaginary-time and thermal-state

algorithms [22, 23, 25] and, with energy gradients in place of trial distances, to the ADAPT-VQE family [26, 27].

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A Implementation and verification details

Reuse of the original implementation. The equations of motion (construction of M and V including the global-phase terms, Tikhonov pseudo-inverse, McLachlan distance, adaptive step, forward Euler), the derivative states, the pools, the models and the exact-diagonalisation initial state are imported unchanged from the authors' package. A two-line shim restores two `numpy` aliases removed in recent `numpy` versions. Two adaptations were made, both verified. (i) *Operator container.* The original stores every pool operator as a dense matrix with its eigendecomposition and forms a dense exponential at every gate application. We substituted an exact Pauli-string container with the same interface, using $e^{\alpha \hat{P}} = \cosh\alpha\, \mathbb{I} + \sinh\alpha\, \hat{P}$ and a signed permutation for $\hat{P}|v\rangle$. (ii) *Selection hook.* The trial-append loop was refactored so that the selection among candidates is a function call. The trial arithmetic is copied verbatim, and without a policy the original sequential rule is replayed over the same candidates in the same order.

Tests. Over 60–300 time steps on each benchmark, the refactored loop without a policy is bit-identical to the authors' class on the same backend: identical operator sequences, identical parameter vectors and identical time grids. Against the original dense backend, run in a separate process without any modification, the Pauli backend reproduces the operator sequences of MFIM $h_z = 0$ and of both LSM runs, with time grids agreeing to 3×10^{-13} and 3×10^{-10}. For MFIM $h_z = 0.5$ the sequences agree for 18 operators and then differ at a decision where several candidates already satisfy $L^2 < L_{\mathrm{cut}}^2$. The original rule then prefers the smallest $\max|\dot{\theta}|$, a quantity set by a pre-existing parameter and equal across these candidates to about 10^{-13}, so round-off decides. The dense Hamiltonian used by the exact reference coincides with the authors' $\hat{H}(t)$ to machine precision on a grid of 61 times. The exact LSM reference changes by at most 2.5×10^{-13} in fidelity when the Magnus step is halved. Inline fidelity tracking agrees with post-hoc evaluation from stored states to 10^{-15}. Re-running seed 0 of the unconstrained study reproduces the first sanity runs bit-for-bit.

Bookkeeping relative to the original driver. (i) The original driver applies the pre-quench Hamiltonian in the first MFIM step (default step 0.01), effectively quenching at $t = 0.01$; we quench at $t = 0$, as in the benchmark definition. (ii) It records post-step states under the pre-step time label; we record them at $t + \delta t$. (iii) It stops at $T + 0.01$; we shorten the last step so that runs end exactly at T (MFIM) and $2T$ (LSM), by lowering the $\delta t_{\max}$ passed to the unchanged step routine.

Compliance audits. Each run directory contains an audit file that lists every two-qubit gate of the compiled circuit: its position, the AVQDS step at which it was added, its kind (rotation or SWAP), its logical and physical qubits, and a flag stating whether the physical pair is a device edge. Section 4 summarises the audits per topology.

B Development-set design selection

Stage 1 screened 150 designs with the authors' pool order on the eight development configurations ($N = 6$ MFIM $h_z \in \{0, 0.5\}$ and LSM $h_z \in \{-0.7, 1.6\}$, each on the 6-qubit heavy-hex tree D6 and the 2×3 lattice G6). Runs were subject to a wall-clock budget of 300 s, 7–20 times a typical development run. All 24 designs with $\epsilon = 1$ exceeded it, and all other designs finished well within it. With $\epsilon = 1$ the window admits every operator with positive gain, so the cheapest one is appended even when its gain is negligible, and the ansatz grows without bound. Of the remaining 126 designs, 110 were feasible. Table 7 lists the best 15. Stage 2 re-evaluated the best ten and the best design of the two other structural families (SWAP-only cost; selection off) with five pool orders (Table 8). The headline design is the first row of Table 8.

Table 7: Stage 1 of the design selection (one pool order, 8 development configurations): the 15 best feasible designs of 150 screened (126 completed within the time budget, 110 feasible).

Design	CX (all)	CX (tree)	CX (grid)	$\Delta f_{\min}$
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, no CAP, Φ_{trial}	0.821	0.992	0.649	-0.0002
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, no CAP, Φ_{trial}	0.833	1.018	0.647	-0.0002
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.827	1.004	0.649	-0.0001
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.846	1.046	0.647	-0.0002
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, CAP, Φ_{unif}	0.867	1.088	0.647	-0.0002
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, CAP, Φ_{unif}	0.867	1.085	0.649	-0.0002
$\epsilon = 0.7$, $\tau = 0$, κ^{cnot}, no CAP, Φ_{trial}	0.881	1.070	0.691	-0.0001
$\epsilon = 0.3$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.879	1.081	0.677	-0.0000
$\epsilon = 0.7$, $\tau = 0$, κ^{cnot}, CAP, no look-ahead	0.882	1.072	0.691	-0.0001
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, no CAP, no look-ahead	0.890	1.132	0.647	-0.0002
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, no CAP, Φ_{unif}	0.894	1.140	0.647	-0.0002
$\epsilon = 0.0$, $\tau = 1$, κ^{cnot}, CAP, no look-ahead	0.894	1.090	0.699	-0.0039
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, no CAP, no look-ahead	0.887	1.125	0.649	-0.0001
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, CAP, no look-ahead	0.887	1.125	0.649	-0.0001
$\epsilon = 0.7$, $\tau = 0$, κ^{cnot}, CAP, Φ_{trial}	0.899	1.106	0.691	-0.0001

Table 8: Stage 2 of the design selection (development set: $N = 6$, D6 tree and G6 lattice, 4 benchmarks, 5 pool orders; 40 runs per design). CNOT and depth multipliers are relative to unconstrained AVQDS of the same pool order; $\Delta f_{\min}$ is the worst final-fidelity change over the 40 runs. The first row is the design selected by the pre-registered rule: the two leading designs both round to 0.89, and the lower mean depth multiplier decides.

Design	CX (all)	CX (tree)	CX (grid)	depth	$\Delta f_{\min}$	feasible
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, no CAP, Φ_{trial}	0.895	1.090	0.699	0.938	-0.0008	yes
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.893	1.087	0.699	0.943	-0.0008	yes
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.915	1.131	0.699	0.977	-0.0008	yes
$\epsilon = 0.7$, $\tau = 1$, κ^{cnot}, CAP, Φ_{unif}	0.919	1.138	0.699	0.990	-0.0008	yes
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, no CAP, Φ_{trial}	0.935	1.170	0.699	0.993	-0.0008	yes
$\epsilon = 0.3$, $\tau = 1$, κ^{cnot}, CAP, Φ_{trial}	0.945	1.147	0.743	0.993	-0.0014	yes
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, CAP, Φ_{unif}	0.945	1.190	0.699	1.021	-0.0008	yes
$\epsilon = 0.7$, $\tau = 0$, κ^{cnot}, no CAP, Φ_{trial}	0.972	1.182	0.762	1.023	-0.0004	yes
$\epsilon = 0.5$, $\tau = 1$, κ^{cnot}, no CAP, no look-ahead	0.968	1.237	0.699	1.029	-0.0008	yes
$\epsilon = 0.7$, $\tau = 0$, κ^{cnot}, CAP, no look-ahead	0.988	1.215	0.762	1.063	-0.0006	yes
$\epsilon = 0.7$, $\tau = 1$, κ^{swap}, CAP, no look-ahead	1.187	1.374	1.000	1.214	-0.0010	yes
selection off, CAP, Φ_{trial}	1.302	1.603	1.000	1.340	$+0.0000$	yes

C All runs

Tables 9 and 10 list every run of the seeded study: 20 unconstrained runs, their 60 naive and 60 SABRE routings, and 60 CA-AVQDS runs on topologies A, B and C. Nothing is omitted; the overhead statistics in the main text are means and standard deviations over exactly these runs. Table 11 shows the breakdown of the naive baseline over insertion strategies and embeddings. The spread over maximum-coverage embeddings is large on B, which is why the headline baseline takes the best combination per circuit rather than an a-priori embedding.

Table 9: N_{cx} of every run (five pool orders per benchmark; seed 0 is the authors' order). Unc.: unconstrained AVQDS (all-to-all); naive: best smart embedding $\times$ insertion strategy; SABRE: reference router on the unconstrained circuit; CA: CA-AVQDS.

			A	B			C		
Benchmark	seed	unc.	CA	naive	SABRE	CA	naive	SABRE	CA
MFIM $h_z = 0$	0	134	84	317	206	173	290	233	207
	1	108	88	264	174	189	243	198	177
	2	106	86	241	160	187	247	184	177
	3	108	90	282	174	168	258	198	189
	4	126	86	294	201	170	252	222	176
MFIM $h_z = 0.5$	0	218	144	542	341	333	476	380	338
	1	208	158	487	331	327	484	400	319
	2	210	152	540	339	309	513	375	317
	3	194	154	470	320	358	470	380	324
	4	194	158	509	323	326	479	368	332
LSM $h_z = -0.7$	0	106	66	181	133	96	106	106	66
	1	90	70	150	108	108	90	90	70
	2	86	66	146	104	91	86	86	66
	3	96	66	150	111	93	96	96	66
	4	96	68	144	114	89	96	96	68
LSM $h_z = 1.6$	0	104	70	182	140	95	104	104	70
	1	92	68	158	119	98	92	92	68
	2	90	66	138	111	118	90	90	66
	3	90	64	156	120	82	90	90	64
	4	104	68	161	131	133	104	104	68

Table 10: Final-time fidelity of every run. Naive and SABRE execute the unconstrained circuit and share its fidelity. N_θ: number of variational parameters.

		fidelity				N_θ			
Benchmark	seed	unc.	CA (A)	CA (B)	CA (C)	unc.	CA (A)	CA (B)	CA (C)
MFIM $h_z = 0$	0	0.9991	0.9994	0.9993	0.9993	124	137	130	142
	1	0.9992	0.9992	0.9991	0.9993	124	151	135	134
	2	0.9993	0.9995	0.9995	0.9994	124	137	136	125
	3	0.9992	0.9994	0.9994	0.9992	122	137	138	141
	4	0.9992	0.9994	0.9993	0.9994	120	125	131	132
MFIM $h_z = 0.5$	0	0.9991	0.9991	0.9992	0.9991	281	281	288	293
	1	0.9992	0.9991	0.9992	0.9992	281	287	287	281
	2	0.9992	0.9991	0.9992	0.9992	292	287	289	291
	3	0.9991	0.9991	0.9992	0.9991	281	284	297	290
	4	0.9991	0.9992	0.9992	0.9991	279	284	284	279
LSM $h_z = -0.7$	0	0.9991	0.9986	0.9985	0.9986	65	95	92	95
	1	0.9993	0.9979	0.9985	0.9979	67	82	99	82
	2	0.9988	0.9988	0.9986	0.9988	67	85	87	85
	3	0.9986	0.9980	0.9982	0.9980	74	88	85	88
	4	0.9986	0.9985	0.9983	0.9985	69	104	99	104
LSM $h_z = 1.6$	0	0.9976	0.9977	0.9982	0.9977	70	98	106	98
	1	0.9984	0.9985	0.9984	0.9985	70	100	94	100
	2	0.9986	0.9985	0.9982	0.9985	70	121	113	121
	3	0.9987	0.9977	0.9984	0.9977	68	109	87	109
	4	0.9989	0.9985	0.9977	0.9985	75	100	145	100

Table 11: Naive baseline on topologies B and C (seed 0): best N_{cx} per shortest-path insertion strategy over all maximum-coverage embeddings, and the distribution of N_{cx} over all (embedding, strategy) combinations. The headline naive baseline is the minimum.

Benchmark	Top.	#emb.	swap-back	move 1st	move 2nd	move both	min	median	max
MFIM $h_z = 0$	B	576	374	329	368	317	317	416	716
MFIM $h_z = 0.5$	B	576	632	578	575	542	542	665	1130
LSM $h_z = -0.7$	B	216	184	199	199	181	181	259	508
LSM $h_z = 1.6$	B	216	182	212	233	203	182	296	464
MFIM $h_z = 0$	C	16	350	290	356	311	290	348	494
MFIM $h_z = 0.5$	C	16	614	476	512	524	476	542	830
LSM $h_z = -0.7$	C	2	106	106	106	106	106	106	106
LSM $h_z = 1.6$	C	2	104	104	104	104	104	104	104

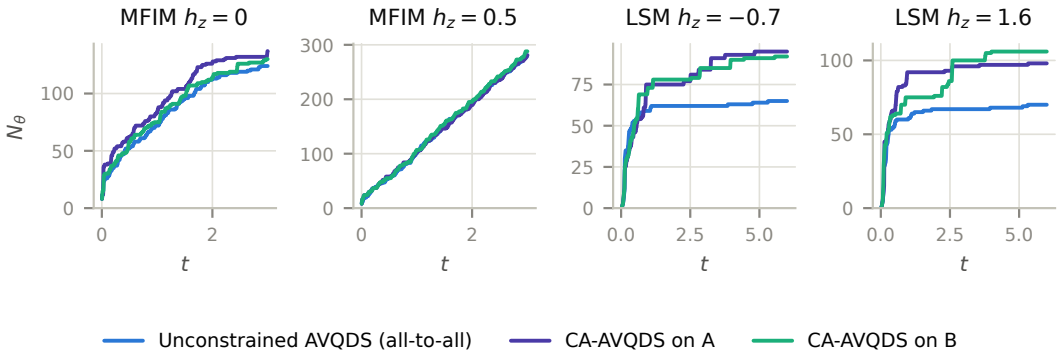


Figure 13: Number of variational parameters $N_\theta(t)$ of unconstrained AVQDS and of CA-AVQDS on A and B (seed 0). The substitution of single-qubit for two-qubit rotations leaves L^2 closer to the threshold and increases N_θ, most for LSM (Sec. 5.2).